

Malavika Ajith

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Summary

B.Tech Computer Science Engineering student with a strong foundation in problem solving, data structures, algorithms, and object-oriented programming. Skilled in Python, C++ and SQL, with hands-on experience building scalable applications and AI/ML-driven solutions through academic projects and internships. Familiar with RDBMS (PostgreSQL, MySQL), NoSQL concepts, and modern web development. Adept at debugging, performance optimization, and technical troubleshooting in fast-paced, collaborative environments. Passionate about delivering robust, efficient, and maintainable software solutions.

Education

Amrita Vishwa Vidyapeetham	Nov 2022 - Present
Bachelor of Technology · Computer Science Engineering · Kollam	7.98/10
Christ Central School	Jun 2021 - Mar 2022
12th Grade · CBSE · Thiruvalla	85.2 %
Chinmaya Vidyalaya	Jun 2019 - Mar 2020
10thGrade · CBSE · Chengannur	95.4%

Experience

Infospica	Aug 2024 - Aug 2024
Web Developer · Web Development Intern · Internship	Trivandrum

Collaborated with the development team to design, build, and optimize web applications focused on user experience, performance, and security. Gained practical experience in front-end development using WordPress, including customization and plugin integration. Learned to implement secure authentication and safeguard user data, strengthening site security protocols.

Hospital Booking Website (WordPress) | Link

- Developed a user-friendly website for Asclepius Hospital to enable efficient appointment booking and easy access to medical services and hospital information.
- Designed and implemented a secure patient portal for accessing medical records and consultation history, ensuring data privacy and compliance with healthcare standards.
- Delivered a platform reflecting the hospital's dedication to quality healthcare through intuitive, accessible UI/UX, robust backend functionality, and code developed to rigorous coding standards.

Projects

AutoPulse-Intelligent Vehicle Lifecycle & Anomaly Management System	July 2025- present
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- Java, Spring Boot, React.js, PostgreSQL, Flask, Isolation Forest, Docker
- Developed a full-stack web application to manage vehicle data and lifecycle tracking with real-time anomaly detection powered by Python-based ML models.
 - Build secure and scalable backend services in Java using Spring Boot, integrating PostgreSQL for persistent storage and role-based access control.
 - Implementing anomaly detection with Isolation Forest via a Flask microservice, triggering "Anomaly Detected" alerts on a live React.js dashboard.
 - Containerize frontend, backend, and ML services using Docker Compose, enabling smooth microservice and environmental consistency.

Deepfake Detection System	Apr 2025 - June 2025
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- PyTorch, Grad-CAM, SHAP, OpenCV, ViT, FaceForensics++, ONNX
- Developed algorithms to detect manipulated video and image content by analyzing subtle visual artifacts and inconsistencies.
 - Utilized state-of-the-art deep learning models (e.g., Vision Transformers) and computer vision techniques to distinguish deepfakes from authentic media.
 - Strengthened multimedia misinformation defense by integrating detection capabilities into broader content verification workflows.

Bias-Aware & Explainable Fake News Detection System	Apr 2025 - June 2025
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- RoBERTa , SHAP, LIME Role: Business Analyst (within Scrum Team)
- Designed and implemented an AI model leveraging deep learning architecture (RoBERTa) for high-accuracy fake news classification.
 - Applied bias mitigation techniques to ensure fairness and reduce model prejudice.
 - Integrated explainability tools like SHAP and LIME to make model decisions transparent and trustworthy.
 - Created comprehensive documentation, including system architecture, algorithm specifications, and procedures for installation and maintenance, facilitating seamless deployment and team collaboration

Certifications

Cyber Security and Ethical Hacking Coincent and Microsoft Took part in a Course in Cyber Security and Ethical Hacking at Coincent (2023)	Mar 2023 -
Evolumin Hackathon ACM Student Chapter Amritapuri National-Level Hackathon (Amrita Vishwa Vidyapeetham, 2024)	Nov 2024 -
CompTIA Network+: Network Services & Data Center Architectures Infosys Springboard Covered network services, data center architectures, and enterprise infrastructure Gained foundational knowledge in network security principles	Mar 2025 -
Problem Solving (Intermediate) HackerRank Demonstrated proficiency in data structures and algorithmic thinking.	Apr 2025 -

Social Outreach Programme

Student Social Responsibility (SSR) Conducted a workshop on Web Development and Career Guidance for Grade 11 students at Lightland Higher Secondary School, Oachira. Conducted a hands-on workshop covering HTML, CSS, and career guidance for higher secondary students.	-
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Technical & Interpersonal Skills

Programming & Scripting Python, C, C++, Java(basics),
Database PostgreSQL,MySQL
Web & Systems Development HTML, CSS, JavaScript, WordPress
Cloud Platforms : AWS (currently enrolled in the Academic Foundation Course)
Leadership & Involvement Leadership,Team Management, Event Coordination
Languages English , Malayalam, Hindi: Fluent to Read, Write, and Speak